



SUJET STATAPP 313 - 2026-2027

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Catégorie : Traitement du signal / NLP

Sujet :

Quantifying and Mitigating Cultural Bias in Language Models

Description du sujet :

Large language models are increasingly used across countries, languages, and cultural contexts, but average benchmark performance can hide substantial disparities between cultural groups. Recent work on cultural evaluation has documented such differences and has also shown that measured cultural alignment can be sensitive to prompt formulation, task format, language, and the choice of cultural proxy. This project addresses two aspects: cultural disparities must first be measured with statistically robust group-level methods, and mitigation methods must then be evaluated not only on average performance but also on their effect on the different groups.

The project is suited to low computational resources. The experiments can be completed using inference on small open-weight instruction-tuned models, stored logits/probabilities, statistical modeling, and inference-time interventions.

Research Question: How can cultural disparities in LLM behavior be measured statistically, and which lightweight debiasing approaches mitigate these disparities?

Project steps:

Step 1: If students are not familiar with large language models (LLMs): the basics of NLP (Transformers, the attention mechanism, LLMs, etc).

Step 2: Get familiar with the benchmark CulturalBench.

Step 3: Evaluate common and small LMs on CulturalBench and interpret the results across different groups. This step implies gathering statistical metrics to measure bias.

Step 4: Try the very low-cost mitigation techniques: prompt sensitivity, multilingual prompting, etc, and compare the results with the previous results.

Step 5: Try at least 2 inference-based techniques and compare the results (methods to adapt accordingly to the students' preference).

Step 6 if there is time: Extend the study to another benchmark.

Données du sujet :

CulturalBench, a human-written and human-verified benchmark designed to evaluate cultural knowledge across 45 countries/regions and a broad range of everyday cultural topics. Paper: <https://aclanthology.org/2025.acl-long.1247/>

If the core study is completed early, students may add a second cultural benchmark or a small manually curated counterfactual set to assess whether conclusions transfer beyond CulturalBench.

Modèles envisagés du sujet :

Open-weight instruction-tuned models that can run on a modest GPU or Google Colab. Suitable examples include Llama-3.2-1B/3B-Instruct and Qwen2.5-1.5B/3B-Instruct.

Bibliographie du sujet :

Adilazuarda, M. F., et al. (2024). Towards Measuring and Modeling ?Culture? in LLMs: A Survey. EMNLP 2024. <https://aclanthology.org/2024.emnlp-main.882/>

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AlKhamissi, B., ElNokrashy, M., AlKhamissi, M., & Diab, M. (2024). Investigating Cultural Alignment of Large Language Models. ACL 2024. <https://aclanthology.org/2024.acl-long.671/>

Wang, Q., Pan, S., Linzen, T., & Black, E. (2025). Multilingual Prompting for Improving LLM Generation Diversity. EMNLP 2025. <https://aclanthology.org/2025.emnlp-main.324/>

Kuzmin, G., Yadav, N., Smirnov, I., Baldwin, T., & Shelmanov, A. (2025). Inference-Time Selective Debiasing to Enhance Fairness in Text Classification Models. NAACL-HLT 2025. <https://aclanthology.org/2025.naacl-short.9/>

Notation du sujet :

Commentaires :